

Divya Gayatri K

AI/ML Engineer

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Summary

AI/ML Engineer with 3 years of experience building Generative AI solutions, RAG pipelines, and intelligent automation systems using Python, LangChain, LLMs, and vector search technologies. Proven ability to develop end-to-end AI workflows for semantic retrieval, context-aware response generation, and data-driven automation across enterprise-scale environments. Strong foundation in machine learning, SQL, and analytics with a focus on delivering scalable AI solutions that improve operational efficiency and business decision-making.

Technical Skills

Generative AI (RAG pipeline development, LLM integration, prompt engineering and optimization)

LangChain (Chains, retrievers, prompt templates, and end-to-end LLM workflow orchestration)

Vector Databases (Embeddings, semantic search, similarity retrieval, and vector indexing)

Data Cleaning & Validation (Data preprocessing, quality analysis, and visualization)

Machine Learning (Regression, Classification, model evaluation, cross-validation, and overfitting mitigation)

Python (Pandas, NumPy, Matplotlib, Seaborn for EDA, ML workflows)

SQL (Data querying, extraction, aggregation, and analytical reporting)

Work Experience

LTM

CIS Engineer

2023–Present

- Contributed to the development of AI-powered RAG applications by building document ingestion, chunking, embedding, and semantic retrieval pipelines for context-aware response generation.
- Improved response relevance and reduced inaccurate AI outputs through prompt engineering, retrieval optimization, and validation testing across multiple query scenarios.
- Worked with large-scale enterprise datasets to perform data validation, reconciliation, and quality analysis, ensuring high data accuracy and operational reliability.
- Developed SQL-based data extraction and analysis workflows to support troubleshooting, reporting, and business-critical operational processes.
- Designed and maintained interactive Power BI dashboards using DAX, Power Query, and data modeling techniques, enabling faster business insights and decision-making.
- Automated repetitive reporting and data validation tasks using SQL and Excel workflows, reducing manual effort and improving operational efficiency.
- Collaborated with cross-functional teams to support analytics initiatives, workflow optimization, and AI-driven automation projects.
- Supported AI experimentation initiatives involving semantic search, vector retrieval, and LLM-based workflows to improve intelligent information access and operational productivity.
- Applied analytical problem-solving and data-driven approaches to identify inconsistencies, improve workflow reliability, and enhance reporting accuracy across systems.

Projects

Response AI – Escalation Analysis Platform (Retail Domain)

Tech Stack – Python, LangChain, RAG, multi-Agent AI, Semantic Retrieval, LLMs, Streamlit, Docker

- Architected a multi-agent AI platform using Retrieval-Augmented Generation (RAG) for automated log ingestion, entity extraction, semantic retrieval, and context-aware issue summarization.
- Built end-to-end AI pipelines integrating specialized agents for orchestration, log analysis, and intelligent response generation, improving escalation resolution efficiency across retail operations.
- Implemented semantic search workflows using embeddings and contextual retrieval strategies, reducing irrelevant outputs through prompt optimization and validation testing.
- Designed modular, scalable AI workflows improving maintainability and operational efficiency across the escalation analysis pipeline.

Q/A Chatbot using RAG (Hospitality Domain)

Tech Stack - Python | LangChain | LLMs | Vector Databases | Semantic Search | Streamlit

- Built an end-to-end RAG-based chatbot for hospitality domain, implementing document ingestion, chunking, embedding generation, vector indexing, and similarity search pipelines.
- Integrated LangChain retrievers, prompt templates, and LLM response chains to orchestrate AI workflows with improved contextual grounding and reduced hallucinations.
- Designed a Streamlit-based interactive UI for chatbot querying and validation; leveraged vector databases for embedding-based semantic retrieval.

Fraud Detection System (Social Sector)

Tech Stack – Python | Pandas | NumPy | Scikit-learn | XGBoost | SMOTE | Matplotlib.

- Developed a machine learning-based Fraud Detection System to identify fraudulent financial transactions, leveraging advanced classification algorithms on highly imbalanced datasets.
- Performed data preprocessing, feature engineering, and class imbalance handling using SMOTE, then trained and optimized Random Forest and XGBoost models to maximize fraud detection recall.
- Evaluated model performance using Precision, Recall, F1-Score, and ROC-AUC, and implemented feature importance analysis to improve model explainability and support risk-based decision-making.

Notification Agent (n8n Automation)

Tech Stack - n8n | Gmail API | WhatsApp API | LLM-based Classification

- Developed an event-driven intelligent automation workflow using n8n Cloud for real-time email monitoring, with LLM-based intent classification to identify interview-related communications.
- Built conditional automation pipelines using n8n IF nodes for intelligent routing; integrated Twilio WhatsApp API for real-time alerting and automated notifications.
- Improved classification accuracy through prompt engineering and multi-scenario testing, reducing manual effort in email monitoring workflows.

Education

Bachelor of Science in Data Science
Aditya Degree College, Kakinada | 2023
CGPA: 8.75

Certifications

2024: Odin School Data Science Professional Certificate – Odin School